ABHISHEK

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SUMMARY

Results-driven Data Scientist with 2 years of experience in leveraging machine learning, deep learning, and statistical methods to tackle high-impact challenges. Expertly developed and deployed predictive models, executed data cleaning and preprocessing tasks, and uncovered actionable insights from extensive datasets. Demonstrated strong analytical acumen and adaptability, consistently driving optimization and innovation in data-driven business solutions.

EXPERIENCE

***DATA SCIENTIST* – ATHENASOFT CLOUD CONSULTING** JAN 2025 – Current

- Architected and deployed a multi-tenant serverless platform using AWS Lambda, API Gateway, DynamoDB, and Cognito, enabling secure self-service onboarding with MFA and OAuth2.0 authentication. Implemented **event-driven Lambda triggers** to automate user-scoped DynamoDB provisioning post-Cognito confirmation, ensuring tenant-level data isolation and horizontal scalability. Designed and secured **RESTful APIs** with JWT-based authorization, granular CORS policies, and least-privilege IAM roles, strengthening API security and compliance. Built global static web delivery architecture using **Amazon S3, CloudFront, Route53, and ACM**, improving latency and enforcing HTTPS across environments. Established **CI/CD pipelines with AWS CDK**, enabling infrastructure-as-code deployments across Cognito, API Gateway, domain, and storage stacks.
- Engineered an **enterprise-grade RAG pipeline** using custom chunking strategies and open-source embedding models, improving retrieval precision and response relevance. Developed an **AI-powered LMS assessment planning bot** using Azure AI Foundry, Blob Storage, and vector stores to generate personalized multi-day learning plans. Implemented **adaptive template regeneration logic** to automatically revise assessment plans when early outputs failed quality thresholds. Built **agentic AI workflows** in Azure AI Foundry (NextGen): Planning agent-generated execution strategy, SQL agent synthesized and executed queries, Vector store provided schema-aware grounding.
- Led development of a **modular time-series forecasting system** (LTC, CfC, CNN) with Optuna-based hyperparameter tuning, improving prediction accuracy by ~12%. Designed a **custom Actor–Critic reinforcement learning pipeline** in Ray RLlib with LTC cores, increasing simulated trading returns by ~15%. Integrated Weights & Biases for experiment tracking, model versioning, and reproducibility. Designed a framework for testing the model performance on **Quant Connect,** including alpha, risk management, and models

Technologies: Python, AWS CDK, Lambda, API Gateway, DynamoDB, Cognito, CloudFront, Route53, S3, IAM, Azure AI Foundry, Semantic Kernel, RAG, Vector Databases, Ray RLlib, Optuna, PyTorch, Node.js, REST APIs, JWT, CORS, Quant Connect.

***DATA SCIENTIST* – TECHGURU** Nov 2023 – Dec 2024

- Designed and deployed a retrieval-augmented chatbot integrated into an LMS platform, enabling curriculum-aligned student assistance. Built **grade-aware assessment planning pipelines** that mapped user queries to academic standards and generated structured learning roadmaps.
- Implemented **dynamic prompt orchestration** to regenerate full assessment templates when initial outputs failed evaluation checks.
- Optimized **semantic search pipelines** using custom chunking and embedding strategies, improving retrieval relevance and response accuracy.
- Integrated AI assistant into production LMS workflows via **REST APIs**, improving student engagement and response turnaround time.
- Developed **response evaluation heuristics** and feedback loops for continuous prompt and retrieval optimization. Collaborated cross-functionally with frontend and backend teams to productionize GenAI features.

Technologies: Python, Azure AI Foundry, RAG, Vector Stores, Prompt Engineering, REST APIs, Blob Storage, Semantic Search, LangChain

***DATA SCIENTIST INTERN* – INEURON INTELLIGENCE PVT. LTD** June 2023 – Oct 2023

- Built an end-to-end credit card default prediction pipeline with automated data preprocessing, feature engineering, and validation workflows, reducing manual preparation time by ~40%.
- Mitigated class imbalance using **SMOTE and undersampling**, and benchmarked multiple ML models with cross-validation and hyperparameter tuning, achieving ~90% accuracy with SVM.
- Implemented **MLOps workflows** including model versioning, reproducible training pipelines, and CI/CD automation; deployed scalable real-time inference on AWS using SageMaker, Lambda, and API Gateway.
- Co-authored an **IEEE research paper on MLOps-driven credit card default prediction**, demonstrating scalable pipeline architecture and model lifecycle automation.

PROJECTS

MULTIMODAL RAG – WHISPER, CLIP, LLAVA AND FAISS *(GitHub Link)*

- Created a system that processes video, audio, and text using CLIP, Whisper, and FAISS for efficient retrieval and indexing.
- Applied Whisper for audio transcription, CLIP for embeddings, and LLaVA/LLaMA models for augmenting and summarizing retrieved data. Built a FAISS-based query engine to retrieve relevant image and text embeddings, enhancing search precision. Automated and streamlined video-to-image, audio-to-text, and embedding processes for multimodal data extraction.

RAG FROM SCRATCH – TECHNOLOGY LLAMA 3 (8B) *(GitHub Link)*

- Developed a GPU-accelerated Retrieval-Augmented Generation (**RAG**) system from scratch using Docker, PyTorch, and Streamlit. Engineered a robust containerized pipeline for scalable document ingestion, embedding, and semantic search. Integrated state-of-the-art NLP models and optimized deployment for CUDA-enabled environments, ensuring high performance and reproducibility.

CREDIT CARD DEFAULTS PREDICTIONS – MLOPS *(GitHub Link)*

- **Architected** and **deployed** a comprehensive end-to-end **MLOps pipeline** on **AWS,** integrating stages of data extraction, validation, preparation, model training, and evaluation, ensuring seamless and automated workflow. **Showcased** collaboration and version control expertise by developing and uploading the project to GitHub, facilitating transparency and teamwork.
- **Streamlined** data management processes and **optimized** model performance through MLOps practices, underscoring machine learning and cloud infrastructure proficiency. **Leveraged** advanced machine learning techniques, including feature engineering and hyperparameter tuning,

to improve prediction accuracy and model robustness. **Demonstrated** strong problem-solving and technical skills by addressing class imbalances and enhancing model generalization through cross-validation.

MULTIPLE DISEASE PREDICTION – TECHNOLOGY STREAMLIT *(GitHub Link)*

- **Developed a scalable machine learning application** using Streamlit, capable of predicting multiple diseases with the flexibility to add more disease detection features in the future. **Implemented multiple models** to enhance prediction accuracy across different diseases, showcasing a robust and modular design. **Demonstrated expertise** in deploying machine learning solutions
- Leveraged **Streamlit** to create an interactive user interface, allowing for real-time disease prediction and easy integration of additional models.

EDUCATION

Master of Science in Data Science – **CHANDIGARH UNIVERSITY** *(MOHALI, PUNJAB)* 2022 – 2024

- Mastered **supervised** and **unsupervised** learning algorithms, including **regression, classification, clustering**, and **neural networks**. Gained hands-on experience in developing, training, and **evaluating models** in dedicated **Machine Learning Deep Learning** Labs using Python, TensorFlow, Keras, and PyTorch. Excelled in data cleaning, preprocessing, and feature engineering, leveraging tools like Pandas and NumPy.
- Applied advanced mathematical concepts, including linear algebra, probability, and statistics, to optimize machine learning algorithms and improve model performance.
- Applied advanced **data mining** techniques and **big data** processing using **Hive**, **HBase**, and **Spark** in Big Data Lab. Analyzed large-scale datasets, developing data-driven strategies to optimize machine learning model performance. Created compelling **data visualizations** using **Matplotlib, Seaborn**, and other tools to communicate insights effectively in the Data Visualization Lab. Developed dashboards and reports to present findings.
- Authored a research paper on **Automated Credit Card Default Prediction Using MLOPs**, which was accepted for publication in **IEEE Xplore journals**, highlighting cutting-edge advancements in Machine Learning and AI. Completed a dissertation on a machine learning project, focusing on real-time deployment and optimization. Developed technical documentation and research papers, with expertise in typesetting using **LaTeX**.

Bachelor of Computer Application – **B.B. S PUBLIC DEGREE COLLEGE** *(PRAYAGRAJ, UP)* 2019 – 2022

- Acquired a strong foundation in programming with **Python, C**, and **C++**, and mastered **data structures and algorithms** through rigorous coursework. Developed software applications and worked on various projects in academic labs.

OTHER

- **Languages**: English, Hindi
- **Technical Skills:**
  - **Programming Languages & Tools:** Python (Pandas, NumPy, Seaborn, Matplotlib, Scikit-Learn), R
  - Machine **Learning & Deep Learning:** Supervised and Unsupervised Learning, Neural Networks, Regression, Classification, Clustering, TensorFlow, Keras, PyTorch, Model Training & Evaluation, Model Fine-Tuning
  - Data **Analysis & Visualization:** Data Cleaning, Preprocessing, Feature Engineering, Data Mining, Data Visualization (Matplotlib, Seaborn), Big Data Processing (Hive, HBase, Spark), Dashboard Creation
  - Advanced Mathematics & Statistics: Probability, Statistical Analysis, Linear Algebra, Operations Research Techniques, Time Series Analysis, Hypothesis Testing
  - Domain-Specific Expertise: Fraud Detection, Location Intelligence, Catalog Enrichment
  - Cloud & MLOps: AWS (Deployment, CI/CD Pipeline), Azure, Docker, Streamlit
  - **Generative** AI: Large Language Models (LLMs), LLaMA 2, Retrieval-Augmented Generation (RAG) Systems
  - Databases & Big Data**:** Hadoop, Database Management Systems (SQL, NoSQL), Big Data Labs
  - Research & Technical Writing: Research Paper Writing, Technical Documentation, IEEE Xplore Publications
  - Other **Tools** & Technologies: Git, Version Control, LangChain
- **Certifications:** ML Specialization, Convolution Neural Network, Intro to ML in Production, Improving Neural Network, Python for Data Science